Lesson 1 — Extra Practice

Polynomials: Review and Preview

Part A — Degree, Coefficients and Order — Pushed Further

1. The degree of a term is the **sum** of the exponents of its variables. State the degree of each monomial. Watch (e): the fraction bar is only dividing by a number.

(a) $-7x^3y^5$ (b) $\frac{2}{9}a^2b^2c^2$ (c) $\sqrt{11}\,m^4n$ (d) $12pq^3r^2s$ (e) $-\frac{u^6v}{5}$ (f) -15

2. For each polynomial in *two* variables, state the degree of every term, then the degree of the whole polynomial, then its name.

(a) $4x^2y - 3xy^3 + x^4$

(b) $7a^3b^2 - a^2b^2 + 5ab - 9$

(c) $m^2n^2 - 6m^3$

(d) $-2p^5q^4$

3. None of these is in descending order, so the leading coefficient is *not* the number you see first. For each: (i) rewrite in descending powers of the variable, then state (ii) the leading coefficient, (iii) the constant term, and (iv) the degree.

(a) $6x - 5 + 2x^4 - x^3$ (b) $-3 + 7y^2 - \frac{1}{2}y^5 + y$ (c) $8t^2 - t^6 + 4t^3$

4. Consider $5xy^3 - 2x^3y + 7x^2y^2 - x^4$. This one is in two variables, which makes "leading coefficient" a question worth asking carefully.

(a) State the degree of each of the four terms, then the degree of the polynomial. What do you notice?

(b) Write the polynomial in descending powers of x, then write it again in descending powers of y.

(c) The booklet defines the leading coefficient for a polynomial in a **single** variable. Explain why that definition does not settle a leading coefficient for this polynomial, and say what answer each of your two orderings in (b) would suggest.

5. Give an example of each. (Answers will vary — but yours must meet *every* condition listed.)

(a) a trinomial in two variables, of degree 5, whose constant term is -4.

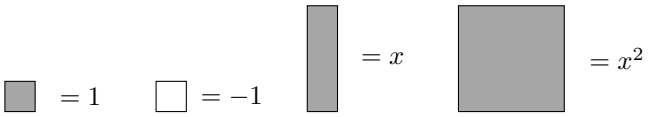
(b) a binomial in one variable, of degree 4, with leading coefficient -3 and no constant term.

(c) a monomial of degree 7 in three variables, with numerical coefficient $\frac{1}{2}$.

(d) a four-term polynomial in one variable, of degree 3, written in descending order, whose four coefficients are four *different* negative integers.

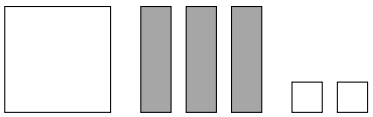
Part B — Algebra Tiles

Tile legend for this course: a **shaded** tile is positive, an **unshaded** (outline only) tile is negative.

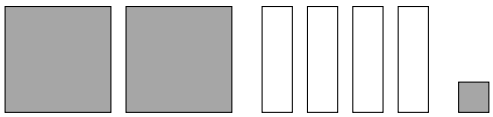


6. State the polynomial expression which describes each diagram. Write each answer in descending powers of x, and be careful in (c) — one of the usual pieces is missing.

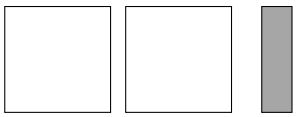
(a)



(b)



(c)



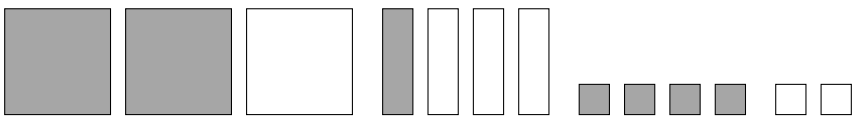
7. Now go the other way. For each polynomial, **draw** the tiles that represent it (shade the positive ones), and state how many tiles you drew altogether.

(a) $3x^2 - 2x + 4$

(b) $-x^2 - 3x$

(c) $2x - 5$

8. The diagram below has not been simplified — it still contains **zero pairs** (a shaded tile and an unshaded tile of the same shape, which together are worth 0).



(a) Write the expression the diagram shows *before* any cancelling — one term for every group of tiles.

(b) How many zero pairs of each shape are there? Say which tiles are left standing.

(c) Write the simplified polynomial in descending powers of x.

9. With tiles, a subtraction is done by **adding the opposite**. Consider $(2x^2 - 3x + 1) - (x^2 - 5x - 2)$.

(a) Write the opposite of $x^2 - 5x - 2$, and say which tiles you would add.

(b) After the two collections are put together, how many zero pairs of each shape form?

(c) State the result, and check it by substituting $x = 1$ into the original expression and into your answer.

Part C — Like Terms in Two Variables

10. Terms are **like** only if they have the same variables raised to the same exponents. For each pair, state like or unlike, and give the reason in terms of exponents.

(a) $3xy$ and $3x^2y$

(b) $-5a^2b^3$ and $7b^3a^2$

(c) $\frac{1}{2}m^2n$ and $\frac{1}{2}mn^2$

(d) $6x^2y$ and $-x^2y$

(e) $4pq$ and $4p$

(f) -9 and $\frac{7}{3}$

11. Simplify by collecting like terms. Sort the terms into groups *before* you add anything — in these expressions xy, x^2y and xy^2 are three different groups.

(a) $5xy - 3x^2y + 8xy + x^2y$

(b) $7a^2b - 4ab^2 - a^2b + 9ab^2$

(c) $m^2n + 3mn^2 - 5m^2n + mn - 2mn^2$

(d) $2x^2y^2 - 3xy + 4 - x^2y^2 + 3xy - 9$

(e) $6pq^2 - 4p^2q + 2pq - 6pq^2 + 4p^2q$

(f) $\frac{1}{2}u^2v + 3uv^2 - \frac{3}{2}u^2v - uv^2$

Part D — Adding and Subtracting Whole Brackets

12. Every subtraction sign in front of a bracket must reach *every* term inside it. Simplify. In (d) work the inner bracket first.

(a) $(4x^2y - 3xy + 7) + (2xy - x^2y - 5)$

(b) $(6a^3 - 2a + 1) - (a^3 - 5a^2 + 2a - 8)$

(c) $(3m^2 - 7m) - (m^2 - 4) - (2m^2 - 7m + 4)$

(d) $(5y^3 - y) - \left[(2y^3 + 4y^2) - (y^3 - y + 6)\right]$

13. Subtract in the vertical format. Change the sign of every term in the bottom polynomial first, then add down the columns. In (b) one column disappears entirely.

(a)
$$\begin{array}{r} 7x^2 - 4x + 9 \\ -\,(2x^2 - 9x - 3) \\ \hline \end{array}$$

(b)
$$\begin{array}{r} -3x^3 + 5x - 8 \\ -\,(x^3 - 2x^2 + 5x - 8) \\ \hline \end{array}$$

(c)
$$\begin{array}{r} 6a^2b - 5ab + 2 \\ -\,(-a^2b + 5ab - 7) \\ \hline \end{array}$$

14. Subtract as directed. Read the wording carefully — "subtract A from B" means $B - A$.

(a) Subtract $(3x^2 - 8x + 4)$ from the sum of $(x^2 + 5x - 6)$ and $(4x^2 - 2x + 1)$.

(b) From $(9m^2n - 4mn + 3)$ subtract the sum of $(2m^2n + mn)$ and $(m^2n - 5mn - 3)$.

(c) What must be *added* to $(2x^2 - 5x + 1)$ to give $(x^2 + 3x - 4)$?

Part E — Find the Error

15. Asked to simplify $2x^2 + 3x$, a student wrote $2x^2 + 3x = 5x^3$.

(a) Substitute $x = 2$ into the original expression and into the student's answer to show the answer is wrong.

(b) Use the definition of like terms to explain why $2x^2$ and $3x$ cannot be combined at all, and state the simplest form of $2x^2 + 3x$.

(c) The same student then wrote $2x^2 + 3x^2 = 5x^4$. Correct it, and say in one sentence what happens to the coefficients and what happens to the exponents when like terms *are* combined.

16. A second student was asked to simplify $(6y^2 - 4y + 5) - (2y^2 - 7y + 3)$ and wrote:

$$(6y^2 - 4y + 5) - (2y^2 - 7y + 3) = 6y^2 - 4y + 5 - 2y^2 - 7y + 3 = 4y^2 - 11y + 8.$$

(a) Identify exactly which terms the student changed correctly and which they did not, and name the mistake in one sentence.

(b) Redo the subtraction correctly by first writing the opposite of the whole second bracket.

(c) Substitute $y = 1$ into the original expression, into the student's answer and into yours. Which two agree, and what does that prove?

Part F — Reasoning About Degree

17. Answer each question with a reason, not just a yes or a no.

(a) Can a **trinomial** have degree 1? If it can, give one; if it cannot, explain why not. Does your answer change if the trinomial is allowed only one variable?

(b) What is the degree of a non-zero constant such as -15? Explain using the rule "the degree of a term is the sum of the exponents of its variables".

(c) Your booklet states that the degree of the polynomial 0 is 0. Using that, state the degree of the answer you found in Question 12(c).

(d) Can a **binomial** have degree 0? Explain what would have to be true of its two terms, and why that cannot happen.

Part G — Stretch

18. A quadrilateral has a perimeter of $(12x^2 + 5x - 7)$ cm. Three of its sides measure $(3x^2 - 2x + 1)$ cm, $(4x^2 + x - 5)$ cm and $(x^2 + 6x)$ cm.

(a) Write and simplify an expression for the length of the fourth side. Name the polynomial you get and state its degree.

(b) Determine all four side lengths when $x = 2$, and check that they add to the perimeter.

(c) Explain why $x = 0$ cannot be an actual value in this situation. Use the side lengths to justify your answer.

19. Build a polynomial to order — this time every condition must hold at once.

(a) Write one polynomial that is a trinomial in x and y, has degree 6, has exactly one term of degree 6 and that term contains *both* variables, has a third term of degree 3, and has 9 as its constant term.

(b) Is your answer the only one possible? Explain. Then show that no polynomial can be a binomial of degree 4 in *one* variable whose two terms *both* have degree 4.

20. *(Multiple Choice)* Which expression simplifies to $-x^2 + 4x - 3$?
A. $(2x^2 + x - 1) - (3x^2 - 3x + 2)$
B. $(2x^2 + x - 1) + (3x^2 - 3x + 2)$
C. $(x^2 + 4x) - (2x^2 - 3)$
D. $(3x^2 - 2x + 5) - (2x^2 - 6x + 8)$

21. *(Numerical Response)* The trinomial $6x^2y^3 - 4xy + 11$ has degree a. The polynomial $4x^5 - 9x^2 + x^7 - 3$ has degree b and, once written in descending order, leading coefficient c. The value of $a \times b \times c$ is _____.
(Record your answer as a three-digit number.)

Lesson 2 — Extra Practice

Multiplying Monomials and Binomials

Part A — Four-Cell Grids, Forwards and Backwards

1. Fill in the four cells of each grid, then add the cells and collect like terms to state the product in expanded form.

(a) $(3x+4)(2x-5)$

	$3x$	$+4$
$2x$		
-5		

(b) $(4m-3n)(2m+5n)$

	$4m$	$-3n$
$2m$		
$+5n$		

(c) $(x^2+6)(2x^2-7)$

	x^2	$+6$
$2x^2$		
-7		

2. Each grid below has already been filled in, but the side lengths have been erased. Write the two binomials that were multiplied, then state the product in expanded form.

(a)

	$6x^2$	$4x$
	$-15x$	-10

(b)

	$10a^2$	$-6ab$
	$35ab$	$-21b^2$

(c)

	$3x^4$	$-12x^2$
	$8x^2$	-32

3. These grids are part-filled. Recover every missing entry — both the missing side lengths and the missing cells — then state the product.

(a)

	$2x$	
	$8x^2$	$28x$
-3		-21

(b)

		$-2q$
$5p$	$15p^2$	
	$12pq$	$-8q^2$

(c) In part (a) you found a missing cell from the two side lengths, and a missing side length from a cell. Explain which of those two jobs is the harder one, and why a grid can sometimes have more than one cell missing and still be solvable.

Part B — Monomials over Several Variables

4. Expand. Every monomial here carries a negative sign, more than one variable, or both — distribute the whole monomial, sign included, to each term inside the bracket.

(a) $-3x^2y(4x - 5y + 2xy)$

(b) $-2ab^2(3a^2 - ab + 4b)$

(c) $5m^2n(-2m + 3n - mn)$

(d) $-pq(p^2 - 4pq - 7q^2)$

(e) $-4c^2d^3(2cd - 3c^2 + d)$

(f) $(3x - 2y - 5z)(-6xy)$

5. Expand and simplify. In each one a **negative** monomial multiplies a bracket that already contains a subtraction — the two signs together decide the sign of the term.

(a) $-2a(3a - 4b) - 5b(2a + b)$

(b) $4xy(2x - 3y) - 3x(2xy - y^2)$

(c) $-x^2(x - 2y) - (3x^3 - x^2y)$

(d) $5mn(m - 2n) - 2m(3mn - n^2)$

(e) $-2uv(3u - v) - 4u(v^2 - 2uv)$

(f) $-3rs^2(2r - s) - rs(s^2 - 6rs)$

(g) Part (f) collapses to a single term. Which two terms cancelled, and what was it about the two monomials out front that made that happen?

Part C — Binomial Times Binomial

6. Expand and simplify.

(a) $(x - 5)(x - 3)$

(b) $(2x + 3)(2x - 3)$

(c) $(4p - 3q)(2p - 5q)$

(d) $(x^2 + 3y)(x^2 - 8y)$

(e) $(6 - 5t)(6 + 5t)$

(f) $(3a - 2b)^2$

7. Each of the four products below is a sum times a difference of the *same* two terms.
(a) $(2x+3)(2x-3)$ (b) $(5y-4)(5y+4)$ (c) $(a+7b)(a-7b)$ (d) $(3m^2-8)(3m^2+8)$

(e) Every answer above is a binomial, not a trinomial. Using FOIL, say exactly which two of the four terms disappear and why they must always cancel.

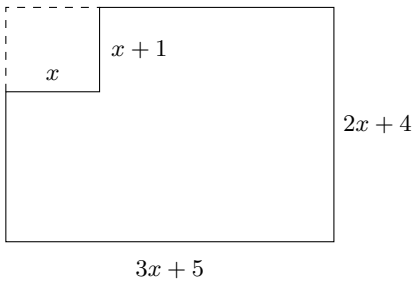
(f) Use the pattern — not a calculator — to evaluate 98×102. State the two terms you are squaring, then the answer.

8. Expand and simplify. Expand *every* bracket before you collect anything.
(a) $(x+4)(x-2)-(x+3)(x-5)$ (b) $3(2x-1)(x+5)$ (c) $(x+3)^2-(x-3)^2$

(d) $(2a+b)(a-3b)+(a+2b)(3a-b)$ (e) $(x-5)(x-3)-(x-5)(x+3)$ (f) $-2(3y-4)(y+2)$

Part D — Composite Areas, Two Ways

9. The shape below is a rectangle $(3x+5)$ wide and $(2x+4)$ tall with a corner piece x by $(x+1)$ removed; the dashed lines show the missing corner.



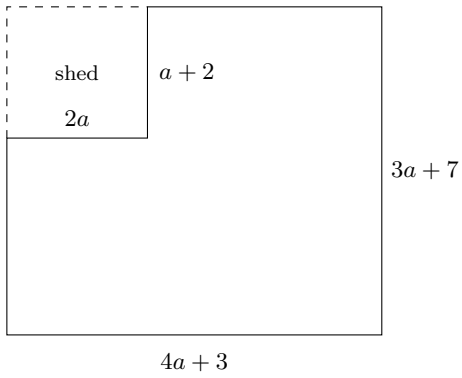
(a) **Whole minus piece.** Write the area as the big rectangle less the corner, then expand and simplify.

(b) **Sum of parts.** Cut the shape with one horizontal line into two rectangles. You will have to work out two of the four side lengths yourself. Write the area as a sum of two products, then expand and simplify.

(c) Your two answers must agree. Verify them a third way by substituting $x = 2$: find the real dimensions of the shape, work out its area arithmetically, and check that both expressions give the same number.

(d) Method (a) needed two multiplications; method (b) needed two multiplications *and* two subtractions first. Say which method you would choose if the cut-out were in the middle of the rectangle instead of at a corner, and why.

10. A rectangular lot is $(4a + 3)$ m by $(3a + 7)$ m. A rectangular shed $2a$ m by $(a + 2)$ m sits in one corner. The lawn is everything outside the shed.



(a) Find the lawn area as the whole lot minus the shed.

(b) Find it again by cutting the lawn with one **horizontal** line into two rectangles.

(c) Find it a third time by cutting the lawn with one **vertical** line into two rectangles.

(d) All three expressions must be identical. One of the pieces cut in (b) or (c) has a monomial side, so its area takes a single distribution rather than a binomial-times-binomial expansion — name it, and say what that tells you about choosing where to cut.

Part E — Working Backwards

11. One factor is missing. Determine it. (Check by expanding.)

(a) $(x+3)(_____) = x^2 + 11x + 24$

(b) $(2x - 1)(_____) = 6x^2 + x - 2$

(c) $(_____)(x + 7) = x^2 - 49$

(d) $(4a + 5b)(_____) = 4a^2 - 3ab - 10b^2$

(e) $(_____)^2 = 9m^2 - 24m + 16$

(f) $(5x + 2)(_____) = 5x^2 - 28x - 12$

12. Determine the value of each unknown coefficient. Every unknown stands for a whole number unless stated otherwise.

(a) $(3x + k)(x - 4) = 3x^2 - 7x - 20$

(b) $(kx + 3)(2x - 5) = 8x^2 - 14x - 15$

(c) $(x + k)(x - k) = x^2 - 49$ (two answers)

(d) $(2a - kb)(2a + kb) = 4a^2 - 25b^2$, $k > 0$

(e) In $(x + 4)(x + c)$ the coefficient of x is 9. Find c and the constant term.

(f) $(2x + p)(3x + q) = 6x^2 + 19x + 15$, p and q natural numbers.

Part F — Find the Error

13. Asked to expand $(2x + 5)(3x - 4)$, a student wrote

$$(2x + 5)(3x - 4) = 6x^2 - 20.$$

(a) Substitute $x = 1$ into the original product and into the student's answer to show the answer is wrong.

(b) Name precisely which of the four FOIL products the student found and which two they never wrote down.

(c) Expand correctly. Then draw the four-cell grid for this product and mark the two cells the student left empty.

14. Asked to expand $(x - 5)(x - 3)$, a second student wrote

$$(x - 5)(x - 3) = x^2 - 3x - 5x - 15 = x^2 - 8x - 15.$$

(a) The middle term $-8x$ is correct. State precisely which single product the student got wrong, and what the correct value of that product is.

(b) Substitute $x = 0$ into the original product and into the student's answer. Explain why $x = 0$ is an especially good value for catching this particular slip.

(c) Expand correctly, and write the one-sentence rule that would have prevented the error.

Part G — Stretch

15. A product of three binomials. Multiply two of them first, then distribute the third binomial over the trinomial you get, one term at a time.

(a) $(x+1)(x+2)(x+3)$

(b) $(2x-1)(x+3)(x-2)$

(c) $(x-4)(x+4)(x^2+16)$ — choose your pairing so the work is short, and say why that pairing was the short one.

(d) In part (a) each factor has degree 1 and the answer has degree 3. Predict the degree of $(x+1)(x+2)(x+3)(x+4)$ and its leading coefficient without expanding, and say how you know.

16. A difference of two squared binomials.

(a) Expand $(a+b)^2$ and $(a-b)^2$ separately, then subtract to prove that $(a+b)^2-(a-b)^2=4ab$.

(b) Prove it a second way: treat the left side as a difference of squares, so that it is (sum of the two binomials)×(difference of the two binomials). Simplify each bracket first.

(c) Use the identity to evaluate 53^2-47^2 mentally. State your a and b, then the answer.

(d) The identity says $(a+b)^2-(a-b)^2$ never contains an a^2 or a b^2 term. Explain, in terms of the middle terms of the two expansions, why the two squared terms are guaranteed to cancel while the middle terms are guaranteed not to.

17. *(Multiple Choice)* Expand $(3x-4y)(2x+5y)$.

A. $6x^2-7xy-20y^2$ B. $6x^2+7xy-20y^2$

C. $6x^2+23xy-20y^2$ D. $6x^2-20y^2$

18. *(Numerical Response)* The expression $(2x+5)(3x-1)-(x+4)(x-6)$ can be written in the form ax^2+bx+c. The value of $a+b+c$ is _____.

(Record your answer as a three-digit number.)

Lesson 3 — Extra Practice

Multiplying Polynomials I: Special Products

Part A — Both Patterns, Pushed Further

1. Square each binomial using $(a \pm b)^2 = a^2 \pm 2ab + b^2$. The "first term" and the "last term" are no longer single letters — square the *whole* term, coefficient and exponent together. Write every answer in descending order.

(a) $(3x^2 - 5y)^2$ (b) $(7a^3 + 2b^2)^2$ (c) $(4m^2 + 9n^3)^2$

(d) $(x^4 - 10)^2$ (e) $(-2p - 5q^2)^2$ (f) $\left(\frac{1}{2}x - 6\right)^2$

2. Each of these is a sum times a difference of the *same* two terms, so the middle terms cancel and only $a^2 - b^2$ survives. Identify a and b first, then square each.

(a) $(2x^3 - 7y)(2x^3 + 7y)$ (b) $(9a^2 + 4b^5)(9a^2 - 4b^5)$ (c) $(10 - 3m^4)(10 + 3m^4)$

(d) $\left(\frac{2}{3}k + 5\right)\left(\frac{2}{3}k - 5\right)$ (e) $(-5c + 8d^2)(-5c - 8d^2)$ (f) $(x^2y - 6)(x^2y + 6)$

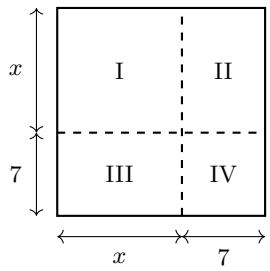
3. **Decide before you expand.** For each product, first write one of **square**, **sum × difference** or **neither**; then expand it — by the matching shortcut if there is one, by FOIL if there is not. (Two of these are disguised: the terms are not written in the order you expect.)

(a) $(4x - 9)(4x + 9)$ (b) $(4x - 9)(4x - 9)$ (c) $(4x - 9)(9x + 4)$

(d) $(3m + 8n)(3m - 8n)$ (e) $(3m + 8n)(8n - 3m)$ (f) $(5p - 2)(5p + 3)$

Part B — Where the Middle Term Comes From

4. The square below has side length $x + 7$, cut into four pieces by the dashed lines.



(a) Write the area of each of the four pieces I, II, III and IV, then add them to give the expanded form of $(x+7)^2$.

(b) Exactly two pieces contribute to the middle term. Name them, and explain why the middle term carries a factor of 2 — that is, why those two pieces are always congruent.

(c) A student claims $(x + 7)^2 = x^2 + 49$. Which pieces of the diagram has the student counted, and which has the student thrown away?

(d) Take $x = 3$. Find the true area of the whole square, find the student's total, and show that the missing area is exactly the area of the two pieces from part (b).

5. The statement $(a + b)^2 = a^2 + b^2$ is the single most common error in this lesson. This question measures exactly how wrong it is.

(a) Evaluate $(a + b)^2$ and $a^2 + b^2$ for $a = 2$, $b = 5$, and again for $a = 6$, $b = 1$. Record the gap between them each time.

(b) Now try $a = -3$, $b = 4$. This time $(a + b)^2$ is the *smaller* of the two. Explain what changed.

(c) Subtract $a^2 + b^2$ from the correct expansion of $(a + b)^2$ to get a formula for the gap. Use it to say exactly when the two expressions *are* equal.

Part C — Find and Fix the Error

6. Asked to expand $(2x - 7)^2$, a student wrote $(2x - 7)^2 = 4x^2 - 49$.

(a) Substitute $x = 1$ into $(2x - 7)^2$ and into the student's answer to show the statement is false.

(b) Expand $(2x - 7)^2$ correctly and state the term the student lost.

(c) The student's answer is not nonsense — it is the correct expansion of a *different* product. Write that product, and say which shortcut the student reached for by mistake.

7. A second student keeps the middle term but gets its sign wrong: $(5m - 3n)^2 = 25m^2 + 30mn + 9n^2$.

(a) Substitute $m = 1$, $n = 1$ into both sides. How far apart are they?

(b) Give the correct expansion, and name the one term that had to change.

(c) Complete the rule: "In $(a \pm b)^2$ the last term is *always* positive because _____, while the middle term takes _____."

8. Four more slips. For each, name the mistake in a few words and give the correct expansion.

(a) $(x + 6)^2 = x^2 + 6x + 36$

(b) $(3y - 4)^2 = 3y^2 - 24y + 16$

(c) $(2a + 5)(2a - 5) = 4a^2 + 25$

(d) $(a^2 - b^2)^2 = a^4 - b^4$

Part D — The Patterns in Reverse: Arithmetic Without a Calculator

9. Every one of these products is a sum times a difference in disguise. Rewrite each in the form $(N + d)(N - d)$ for a round number N, then evaluate it as $N^2 - d^2$. Show the two squares.

(a) 51×49

(b) 88×92

(c) 37×43

(d) 75×85

(e) 19×21

(f) 995×1005

10. Now the squaring pattern. Write each number as $N \pm d$ with N round, then use $(N \pm d)^2 = N^2 \pm 2Nd + d^2$. Write the three pieces before you add them.

(a) 102^2

(b) 49^2

(c) 205^2

(d) 998^2

(e) 31^2

(f) 9.8^2

11. When is the sum-times-difference trick actually worth using?

(a) 47×53 works beautifully. State the condition two factors must satisfy for this method to apply at all.

(b) Use it on 64×56.

(c) Try it on 64×58. The method still works — explain why it saves you nothing here.

(d) Use it on 203×197.

Part E — Adding Two Squares: What Survives

12. Every one of these *adds* a pair of squared binomials. Expand each fully, then collect, and watch what disappears.

(a) $(x+5)^2 + (x-5)^2$

(b) $(a+b)^2 + (a-b)^2$

(c) $(2m+5n)^2 + (2m-5n)^2$

(d) $(4p-q)^2 + (4p+q)^2$

(e) $(x+y)^2 - (x^2+y^2)$

(f) $(a+3)^2 + (a-3)^2 - 2a^2$

13. Read your answers to question 12 back.

(a) In 12(a)–(d) every middle term vanished. Say why, and write the general result for $(a+b)^2 + (a-b)^2$.

(b) The matching identity for a *difference* is $(a+b)^2 - (a-b)^2 = 4ab$. In one sentence, say why adding destroys the cross terms while subtracting destroys the squares instead.

(c) Use your identity from 13(a) to evaluate $27^2 + 23^2$ mentally. State your a and b first.

(d) Without expanding, write down the value of $(6x+7)^2 + (6x-7)^2$.

14. A coefficient out front, and two different patterns in the same expression. Expand every product completely before collecting.

(a) $3(x+4)^2 - (x-4)(x+4)$

(b) $(2a-3b)^2 + (2a+3b)(2a-3b)$

(c) $5(m+2n)(m-2n) - 2(m-2n)^2$

(d) $(x^2+1)(x^2-1) - (x^2-1)^2$

Part F — Stretch

15. Squaring a *trinomial*. The square pattern still does all the work — you just have to use it twice, by treating $(a+b+c)^2$ as the square of the two terms $(a+b)$ and c.

(a) Expand $(a+b+c)^2$ completely that way, showing the step before you expand $(a+b)^2$.

(b) State the result as a rule in words: how many squared terms, how many cross terms, and what coefficient does each cross term carry?

(c) Check your rule with $a=1$, $b=2$, $c=3$ by evaluating both $(a+b+c)^2$ and your expansion.

(d) Use the rule to expand $(2x-3y+5)^2$ in one pass. Write the answer with the squared terms first.

16. A difference of squares in which each "square" is itself a binomial. Group the terms into an a and a b first, apply a^2-b^2, and only then expand what is left.

(a) $(x+y+4)(x+y-4)$

(b) $(3a+b-2)(3a+b+2)$

(c) $(m-n+5)(m+n-5)$ (Hint: group $-n+5$ as $-(n-5)$.)

(d) $(3x+2y)^2-(x-4y)^2$ (Do it as a difference of squares, not by expanding both.)

17. Chaining the difference of squares.

(a) Expand $(x-y)(x+y)(x^2+y^2)$ by multiplying the first two brackets first.

(b) Extend the idea: expand $(x-1)(x+1)(x^2+1)(x^4+1)$.

(c) Put $x=3$ in your answer to (b) to evaluate 3^8-1 from four small factors. Verify the two numbers agree.

18. *(Multiple Choice)* The expansion of $(4x^2-7y)^2$ is

A. $16x^4-49y^2$ B. $16x^4+49y^2$
C. $16x^4-56x^2y+49y^2$
D. $8x^4-56x^2y+49y^2$

19. *(Numerical Response)* When $(3m+4n+2)^2$ is expanded and simplified, the coefficient of the mn term is _____.
(Record your answer as a three-digit number.)

Lesson 4 — Extra Practice

Multiplying a Binomial by a Trinomial, and Three Binomials

Part A — Six Partial Products, Every Time

1. Not one of these trinomials is written in descending order. **Reorder the trinomial first**, then multiply. A binomial times a trinomial is $2 \times 3 = 6$ partial products — count them before you collect. Write every answer in descending order.

(a) $(2x - 5)\left(4x - x^2 + 3\right)$

(b) $(3a + 2b)\left(5a^2 - 4b^2 + ab\right)$

(c) $(7 - 2m)\left(m^2 - 6 + 5m\right)$

(d) $(4p - 3q)\left(2q^2 + 9p^2 - 5pq\right)$

(e) $\left(x^2 + 3\right)\left(2x - 5x^3 + 8\right)$

(f) $(-3y + 2)\left(6 - y^2 - 4y\right)$

2. Now the **vertical** layout, which lines like terms up in columns so nothing can go missing. Write the trinomial on top and the binomial beneath it, multiply the whole trinomial by each term of the binomial on its own line, keeping like terms in the same column, then add down. The skeleton for (a) is shown.

$$\begin{array}{r} x^2 - 3x + 5 \\ 2x - 7 \\ \hline \end{array}$$

row 1: $-7\left(x^2 - 3x + 5\right)$
row 2: $2x\left(x^2 - 3x + 5\right)$
then add the columns

(a) $\left(x^2 - 3x + 5\right)(2x - 7)$

(b) $\left(2a^2 + a - 6\right)(3a + 4)$

(c) $\left(m^2 - 4mn + n^2\right)(m + 3n)$

(d) $\left(5 - 2k + k^2\right)(k - 3)$

3. Expand every bracket *completely* before you collect anything. One of these needs nine partial products, not six.
(a) $2(x-4)\left(x^2+3x-5\right)$ (b) $(2y+3)\left(y^2-y+4\right)-(y-1)(y+1)$

(c) $\left(a^2-2a+3\right)\left(a^2+a-1\right)$ (d) $5x(x-2)\left(x^2+2x+4\right)$

Part B — Three Binomials: Choose Your Pairing

4. Multiplication is commutative, so *any* two of the three brackets may be multiplied first — but the three choices are not equally much work. For each product, **name the pair you would multiply first**, say in one sentence why that pair is the cheap one, then finish the expansion.
(a) $(x+2)(x-2)(x+5)$ (b) $(3x-1)(2x+7)(2x-7)$ (c) $(a+4)(a-3)(a-4)$

(d) Expand 4(a) a second time, this time multiplying the *second* and *third* brackets first. Confirm you get the same cubic, and state how many terms you had to carry through the middle step each way.

5. No shortcut available here — two of the three brackets must be FOILed, and the trinomial that results is then multiplied by the third binomial. Usually that means six partial products in the second stage — but watch for a first pair whose product is only a binomial.
(a) $(2x-3)(x+4)(3x-1)$ (b) $(a-2b)(a+2b)(a+3b)$ (c) $(y-5)(y-2)(2y+1)$

Part C — Cubing a Binomial

6. A cube is a square times one more copy. Build $(x+3)^3$ in two clean stages, then do the same for the rest.

(a) Expand $(x+3)^2$, then multiply that trinomial by $(x+3)$ to get $(x+3)^3$.

(b) $(x-2)^3$

(c) $(2x+1)^3$

(d) $(a+b)^3$, keeping a and b as letters.

7. Look back at your four answers to question 6.

(a) Every expansion had four terms with coefficients in the same pattern. Write that pattern, and describe what happens to the exponent of a and the exponent of b as you read $(a+b)^3$ left to right.

(b) From 6(b), state what changes when the binomial is a *difference*, $(a-b)^3$. Which terms turn negative, and why those ones?

(c) Use the pattern — do not multiply out in stages — to write $(x+5)^3$ and $(3m-2n)^3$.

(d) Write $(x-4)^3$ from the pattern, then check your expansion by substituting $x=1$ into both $(x-4)^3$ and your four terms.

(e) The coefficients of $(a+b)^2$ are $1, 2, 1$ and those of $(a+b)^3$ are $1, 3, 3, 1$. Predict the five coefficients of $(a+b)^4$, and say how you got them from the row above.

Part D — Working Backwards from a Cubic

8. One factor has been erased. Determine it, and check by expanding. (In each case the missing factor has whole-number coefficients.)

(a) $(x+3)(_____) = x^3 + 7x^2 + 15x + 9$

(b) $(_____)\,(x^2 - 2x + 5) = 2x^3 - 7x^2 + 16x - 15$

(c) $(x-1)(x+4)(_____) = x^3 + 5x^2 + 2x - 8$

(d) $(x+k)^3 = x^3 + 12x^2 + 48x + 64$. Determine k, and say which single term of the expansion you could have used on its own.

9. Determine the value of each unknown coefficient. Every unknown stands for an integer. You do *not* need to expand everything — match one column at a time.

(a) $(x+2)\left(x^2+ax+b\right)=x^3+x^2-8x-12$

(b) In $(3x-2)\left(x^2+5x+k\right)$ the constant term is -14. Determine k, then the full expansion.

(c) In $(2x+a)\left(x^2-3x+4\right)$ the coefficient of x is -1. Determine a, then the full expansion.

(d) $(x+5)\left(x^2+bx+c\right)=x^3+2x^2-13x+10$

Part E — Find and Fix the Error

10. Asked to expand $(2x-3)\left(x^2+4x-5\right)$, a student wrote

$$(2x-3)\left(x^2+4x-5\right)=2x^3+8x^2-3x^2+15=2x^3+5x^2+15.$$

(a) How many partial products should there have been, and how many did the student actually write?

(b) Name the two partial products that were never formed, and state which term of the binomial met which term of the trinomial in each.

(c) Expand correctly, and show that the difference between the correct answer and the student's is exactly the sum of the two lost products.

(d) Substitute $x=1$ into the original product and into the student's answer. Explain why $x=1$ makes this error especially easy to catch.

11. A second student wrote $(x+3)^3=x^3+27$, reasoning that "the cube goes on each term."

(a) Substitute $x=1$ into both sides to show the statement is false, and say how far apart the two values are.

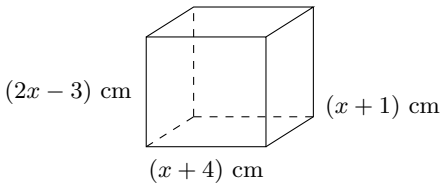
(b) Expand $(x+3)^3$ correctly and name the two terms the student threw away.

(c) The student's answer is the correct expansion of a *different* product: verify that $(x+3)\left(x^2-3x+9\right)=x^3+27$. Compare this second bracket with the one the student needed.

(d) Solve $9x^2+27x=0$ to find the only two values of x for which the student's statement happens to be true.

Part F — Volume

12. A rectangular box has the dimensions shown.



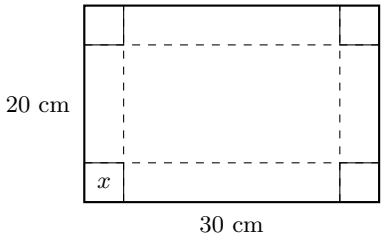
(a) Write the volume as a product of three binomials, then expand it to the form $V = ax^3 + bx^2 + cx + d$.

(b) Evaluate your expansion at $x = 5$, then multiply the three dimensions at $x = 5$ directly. The two results must agree — show both.

(c) Explain why x cannot be 1, and state the smallest whole number x may be.

(d) Which single coefficient of your expansion tells you the volume of a box whose three dimensions are x, x and $2x$ for very large x? Explain.

13. An open-topped box is made from a 30 cm by 20 cm sheet of cardboard by cutting a square of side x cm from each corner and folding the flaps up.



(a) Write the three dimensions of the finished box in terms of x, then write V as a product of three factors.

(b) Expand V and write it in descending order.

(c) Evaluate V when $x = 3$ and again when $x = 4$, both from the expansion and from the three dimensions.

(d) State the values of x that make sense here, and explain what goes wrong at $x = 0$ and at $x = 10$.

Part G — Stretch

14. **Four binomials.** Pair them off before you multiply anything.

(a) $(x+1)(x-1)(x+2)(x-2)$ — choose the pairing that makes two differences of squares, then multiply the two results.

(b) $(2x-1)(2x+1)(x-3)(x+3)$

(c) $(x+1)(x+2)(x+3)(x+4)$ — no shortcut pairing exists, so multiply in two pairs and then multiply the two trinomials.

(d) State the degree and the leading coefficient of the product of any four binomials of the form $(px+q)$, without expanding, and justify both.

15. **Reading a cubic off its factors.** Expand $(x+a)(x+b)(x+c)$ in full, keeping a, b and c as letters.

(a) Show that the expansion is $x^3+(a+b+c)x^2+(ab+ac+bc)x+abc$, and say in words what each of the three coefficients is built from.

(b) *Without expanding*, write down the x^2 coefficient, the x coefficient and the constant of $(x+2)(x-5)(x+9)$. Then expand to check.

(c) Do the same for $(x-1)(x-3)(x-8)$.

(d) Determine the value of c for which $(x+2)(x-7)(x+c)$ has *no* x^2 term, and state the resulting cubic.

16. *(Multiple Choice)* The expansion of $(2x-1)\left(x^2-3x+4\right)$ is

A. $2x^3-7x^2+11x-4$ B. $2x^3-5x^2+11x-4$
C. $2x^3-7x^2+5x-4$ D. $2x^3-6x^2+8x-4$

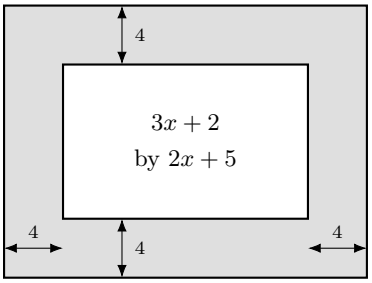
17. *(Numerical Response)* When $(2x+3)^3$ is expanded and simplified, the coefficient of the x^2 term is _____.
(Record your answer as a three-digit number.)

Lesson 5 — Extra Practice

Applying Polynomial Products to Area, Volume, and Equations

Part A — Shaded and Composite Regions

1. A rectangular photograph measuring $(3x+2)$ cm by $(2x+5)$ cm is mounted on a rectangular mat. The mat is a uniform 4 cm wide on all four sides. The shaded region is the mat itself.



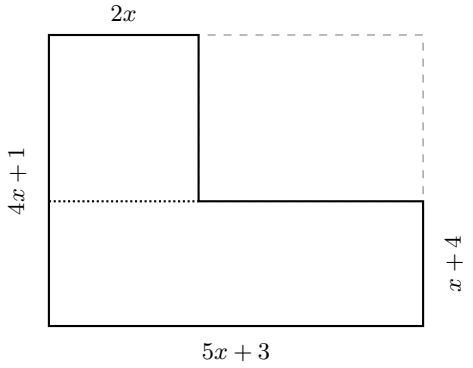
(a) Write the outer length and the outer width of the mat in terms of x. (Each one grows by 4 at *both* ends.)

(b) Expand the outer area and the photograph's area, then subtract to find the shaded area.

(c) Find the shaded area when $x = 5$. Check your answer by working out both real rectangles and subtracting the two numbers.

(d) Your answer to (b) is not a quadratic — the x^2 terms cancelled. Explain what it is about a mat of **constant** width that forces that cancellation.

2. The L-shaped figure below is $(5x+3)$ wide and $(4x+1)$ tall overall. Its upright arm is $2x$ wide and its bottom band is $(x+4)$ tall. The dashed rectangle is the missing corner.

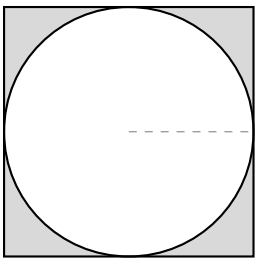


(a) **Add the pieces.** Cut the figure along the dotted line into a bottom rectangle and an upright rectangle. Only one of the upright rectangle's side lengths is printed on the diagram — work out the other first, then write the total area as a sum of two products and simplify.

(b) **Bounding box minus the notch.** Write the area of the whole $(5x+3)$ by $(4x+1)$ box, then the area of the dashed notch. Neither notch side length is printed either. Subtract and simplify. (The notch turns out to be a difference of squares.)

(c) Your two trinomials must match. Check both by putting $x = 2$: draw the figure with real numbers on it and find its area by arithmetic alone.

3. A circle is inscribed in a square of side $(2x + 6)$ cm — it touches all four sides. The shaded region is what is left of the square outside the circle.



$2x + 6$

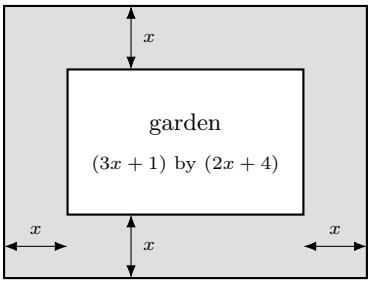
(a) Write the radius of the circle in terms of x.

(b) Write the shaded area, leaving π in your answer. Expand both the square's area and the circle's area; do **not** replace π with a decimal.

(c) Find the shaded area when $x = 2$, to the nearest tenth of a square centimetre.

(d) Write the circle's area divided by the square's area and simplify. What do you notice, and what does it say about how much of *any* such square the inscribed circle covers?

4. A rectangular garden measures $(3x + 1)$ m by $(2x + 4)$ m. A gravel path of uniform width x m is laid all the way around the outside of it. The shaded region is the path.



(a) Write the outer length and the outer width of the gravelled rectangle in terms of x.

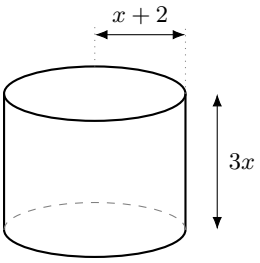
(b) Determine a simplified expression for the area of the path.

(c) Calculate the area of the path when $x = 3$. Check by computing both real rectangles.

(d) In question 1 the x^2 terms cancelled and here they do not. Explain the difference between the two borders that causes it.

Part B — Volume and Surface Area

5. A closed cylindrical can has radius $(x + 2)$ cm and height $3x$ cm. Leave π in every answer — do not replace it with a decimal at any stage.



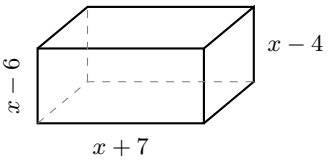
(a) Write and simplify an expression for the volume in cm^3, using $V = \pi r^2 h$. Square the radius binomial first.

(b) Write and simplify an expression for the total surface area in cm^2, using $SA = 2\pi r^2 + 2\pi rh$ — two circular ends plus the curved wall.

(c) State the degree of each answer in x. (π is a number, not a variable, so it does not affect the degree.) Explain why volume has to come out one degree higher than surface area.

(d) Evaluate both when $x = 1$, leaving π in the answers. Check each one against the can's real radius and height.

6. A closed rectangular storage bin has length $(x + 7)$ m, width $(x - 4)$ m and height $(x - 6)$ m.



(a) Write and simplify an expression for the volume in m^3.

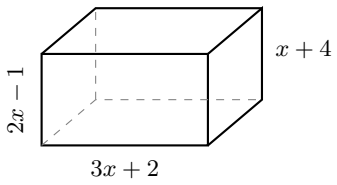
(b) Calculate the volume when $x = 10$, and confirm it by multiplying the three real dimensions.

(c) Now substitute $x = 5$. Write down the number your expression produces.

(d) That answer must be rejected. Work out all three dimensions at $x = 5$ and name precisely the one that makes the bin impossible.

(e) State the full set of values of x for which this bin exists, and give the smallest whole-number value of x together with the bin's dimensions and volume there.

7. A closed rectangular prism has length $(3x + 2)$ cm, width $(x + 4)$ cm and height $(2x - 1)$ cm.



(a) Write and simplify an expression for the volume in cm^3. Multiply two factors first, then distribute the third.

(b) Write and simplify an expression for the surface area in cm^2, using $SA = 2(\ell w + \ell h + wh)$.

(c) State the degree of each of your two answers, and explain why volume comes out one degree higher than surface area no matter what the three side polynomials are.

(d) Evaluate both when $x = 2$. Write down the three real dimensions and confirm both numbers by arithmetic.

Part C — Equations Where the Leading Terms Cancel

8. Expand *both* sides of each equation. In every one the highest-degree terms are identical on the two sides and cancel, leaving a linear equation. Solve it, and give exact fractions where they occur.

(a) $(x + 6)(x - 2) = (x + 3)(x - 1)$

(b) $(2y + 5)^2 = 4y(y + 7)$

(c) $(m - 4)(m + 4) = (m - 2)^2 - 4$

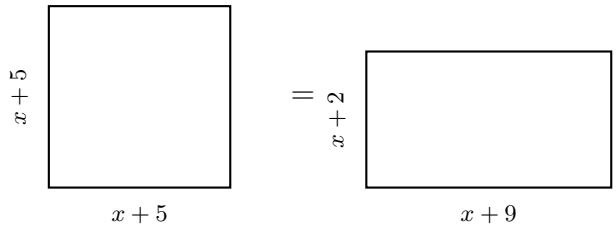
(d) $3t(t + 2) - (t - 1)(3t + 5) = 14$

(e) $(x + 2)^3 = x^3 + 6x^2 + 20$

(f) $(n+1)(n+2)(n+3) = n(n+1)(n+5)$

(g) In parts (e) and (f) *two* degrees cancelled, not one. Say which two, and what has to be true of the two sides for that to happen.

9. A square has side length $(x + 5)$ cm. A rectangle measures $(x + 9)$ cm by $(x + 2)$ cm. The two figures have equal areas.

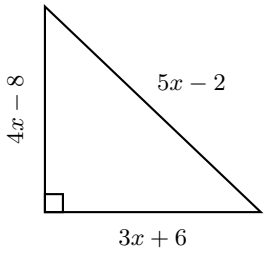


(a) Write an equation, expand both sides, and solve for x.

(b) State the side length of the square and the two side lengths of the rectangle.

(c) Calculate both areas separately and confirm that they agree.

10. In the right triangle below the legs measure $(3x + 6)$ cm and $(4x - 8)$ cm, and the hypotenuse measures $(5x - 2)$ cm.



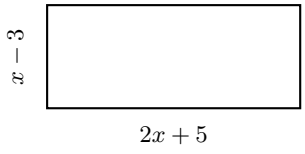
(a) Apply the Pythagorean theorem and expand all three squares. Show that the x^2 terms cancel, then solve for x.

(b) State the three side lengths, and verify the triangle with $a^2 + b^2 = c^2$ using those numbers.

(c) Determine the perimeter and the area of the triangle.

Part D — Evaluating, and When to Reject an Answer

11. A rectangle has length $(2x + 5)$ cm and width $(x - 3)$ cm.



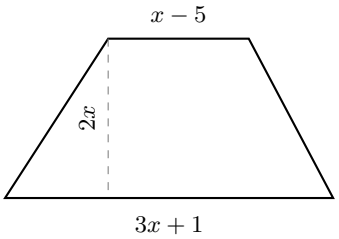
(a) Write and simplify an expression for the area in cm^2.

(b) Calculate the area when $x = 7$, and confirm it against the two real side lengths.

(c) Now substitute $x = 1$. Write down the number your expression produces, then state why that number must be **rejected**. Name the side length that causes the problem and give its value.

(d) State the complete set of values of x for which this rectangle exists, and give the smallest whole-number value of x that works together with the rectangle's dimensions and area at that value.

12. The trapezoid below has parallel sides $(3x + 1)$ cm and $(x - 5)$ cm, and height $2x$ cm. Its area is $A = \frac{1}{2}h(a + b)$.



(a) Write and simplify an expression for the area in cm^2. Add inside the bracket before you multiply.

(b) Calculate the area when $x = 6$, and check it against the three real measurements.

(c) Substitute $x = 2$. State the value your expression gives and why it must be rejected.

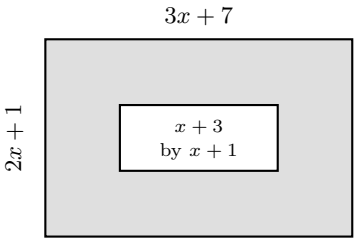
(d) Substitute $x = 5$. This time the expression gives a perfectly ordinary positive number — but the figure is still not a trapezoid. Explain what it has become, and say what the number you calculated is actually the area of.

(e) State the set of values of x for which this really is a trapezoid.

Part E — Find the Error

13. A picture frame has outer dimensions $(3x + 7)$ cm by $(2x + 1)$ cm, with a rectangular opening $(x + 3)$ cm by $(x + 1)$ cm cut out of the middle. Asked for the shaded area, a student wrote

$$2\big[(3x + 7) + (2x + 1)\big] - 2\big[(x + 3) + (x + 1)\big] = 2(5x + 8) - 2(2x + 4) = 6x + 8.$$



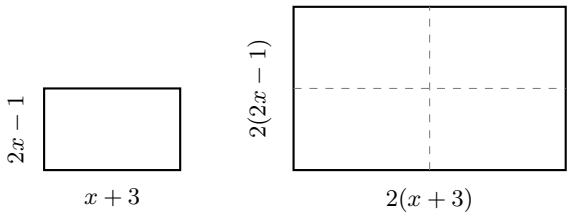
(a) Name the student's mistake in one sentence. What did they subtract instead of what they should have subtracted?

(b) Determine the correct shaded area.

(c) Substitute $x = 2$ into the real figure, the correct expression, and the student's expression. Report all three numbers.

(d) The student's answer has degree 1 and yours has degree 2. Explain how a quick look at the degree alone would have caught this error before any numbers were substituted.

14. A rectangle measures $(x+3)$ cm by $(2x-1)$ cm. A second student claims: "If I double the length and double the width, the area doubles too."



(a) Expand the original area.

(b) Write the two doubled side lengths as polynomials, expand the new area, and write it as a number times your answer to (a).

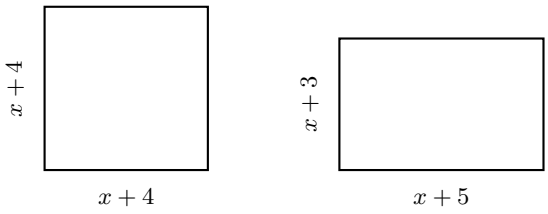
(c) By what factor did the area really change? Test it at $x = 2$ with both sets of real dimensions.

(d) The dashed lines on the second figure show why. Use them to explain the factor in one sentence.

(e) By what factor does area change if every dimension is **tripled**? By what factor does the volume of a box change if every one of its three dimensions is doubled? Explain both.

Part F — Stretch

15. Below are a square of side $(x+4)$ cm and a rectangle measuring $(x+5)$ cm by $(x+3)$ cm. The rectangle is a centimetre longer and a centimetre shorter than the square.



(a) Expand each of the two areas.

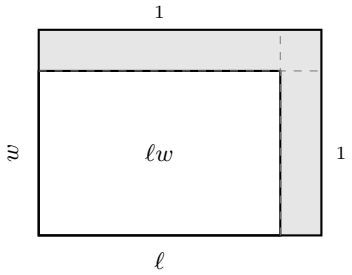
(b) Subtract the rectangle's area from the square's. Simplify completely.

(c) Your answer contains no x at all. Test it at $x = 1$ and at $x = 10$ by working out all four real side lengths.

(d) Write a sentence a classmate could understand explaining what your answer to (b) means about these two figures — and say whether making x enormous would ever let the rectangle catch up.

16. A rectangle has length ℓ and width w. Both the length and the width are increased by 1.

(a) Write the new area as a product, then expand it.



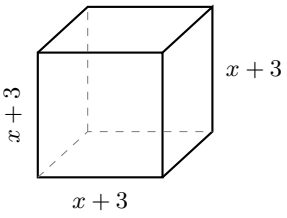
(b) Subtract the original area to find the **increase**, and simplify.

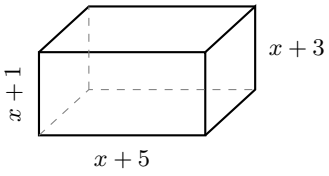
(c) Test your result with $\ell = 7$ and $w = 4$: find both areas and the increase, and check it against your formula.

(d) The shaded part of the figure is exactly the increase. Identify its three pieces and match each one to a term of your answer to (b).

(e) Repeat the whole argument for an increase of 2 in each dimension. Give the increase in area, and check it with $\ell = 7$, $w = 4$.

17. A cube has edge length $(x + 3)$ cm. A rectangular prism has dimensions $(x + 1)$ cm, $(x + 3)$ cm and $(x + 5)$ cm — the same middle measurement, with the other two moved 2 cm apart.





(a) Expand the cube's volume.

(b) Expand the prism's volume. Multiply two factors first, then the third.

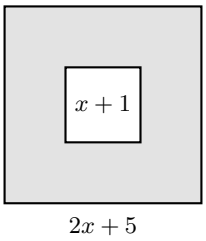
(c) Subtract to find how much larger the cube is. Two whole degrees cancel — name them — and factor what is left.

(d) Check your difference at $x = 2$ by computing both volumes from their real dimensions.

(e) Set your difference equal to zero and solve. Explain why that solution does *not* give a pair of solids with equal volume, and conclude that the cube always wins.

Part G — Multiple Choice and Numerical Response

18. *(Multiple Choice)* A square photograph of side $(x+1)$ cm is mounted centred on a square card of side $(2x+5)$ cm. The shaded area of card still showing is



A. $x^2+8x+16$ **B.** $3x^2+18x+24$ **C.** $3x^2+20x+24$ **D.** $4x+16$

19. *(Numerical Response)* A square of side $(x+7)$ cm has exactly the same area as a rectangle measuring $(x+3)$ cm by $(x+13)$ cm. The value of x is _____. *(Record your answer as a three-digit number.)*

Lesson 6 — Extra Practice

Polynomial Operations Unit Review

Part A — Vocabulary and Structure

1. Three of a polynomial's features — its **degree**, its **leading coefficient** and its **constant term** — can be predicted for a product *before* the product is expanded. This question tests that.

(a) Write a trinomial P in x that has degree 2, leading coefficient 3 and constant term -5. Write it in descending order.

(b) Let $Q = 2x - 7$. State the degree, leading coefficient and constant term of Q, then **predict** all three for the product PQ. Give a reason for each prediction.

(c) Now expand PQ and write the answer in descending order. Confirm that all three of your predictions in (b) were right.

(d) Your prediction for the constant term used only one partial product out of six. Name that partial product, and explain why no other partial product can ever land in the constant term.

2. **Audit each product without fully expanding it where you can.** For each, state (i) the degree of the simplified product, (ii) how many terms it has once simplified, and (iii) its leading coefficient. Two of the six behave unexpectedly — find them.

(a) $(2x - 5)(2x + 5)$

(b) $(x^2 + 3)(x^2 - 4x + 1)$

(c) $(3x - 1)^2 - 9x^2$

(d) $(x + 2)(x - 3)(x + 1)$

(e) $-2x^3y(4xy^2 - 3x^2)$

(f) $(x^2 - y^2)^2$

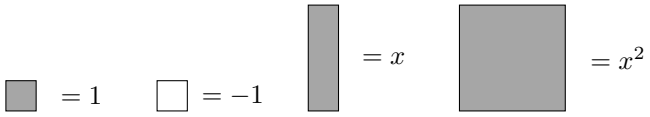
3. Expand and simplify, then write each answer in **descending order**. For each answer also name the polynomial (monomial, binomial, trinomial or just "a polynomial") and state its degree, leading coefficient and constant term. Watch (a) — the trinomial inside the second bracket is not written in order.

(a) $(2y - 3)(y^2 - 5 + 4y)$

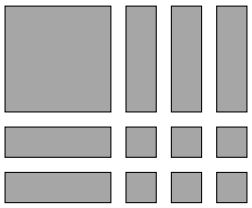
(b) $(1 - 3m)(m^2 + m - 2)$

(c) $(x^2 + 2x)(x^2 - 2x) - x^4$

4. **Tiles and products together.** Tile legend, as in Lesson 1: a **shaded** tile is positive, an **unshaded** (outline only) tile is negative.



The tiles below have been arranged into one large rectangle.



(a) Read the width and the height of the whole rectangle off the tiles, then write the product they represent and its expansion in descending order.

(b) State the degree, leading coefficient and constant term of your answer to (a), and say which group of tiles carries each of the three.

(c) Without drawing anything, say how many tiles of each kind $(x+4)(x+5)$ would need, and state the expansion.

(d) Explain why $(x+2)(x-3)$ cannot be shown as one all-shaded rectangle of tiles. Expand it anyway, and say which tiles would form zero pairs.

Part B — Expanding with Every Tool

5. Each expression needs *more than one* technique: a special product, a binomial times a trinomial, and collecting like terms at the end. Expand every bracket completely before you collect anything, and write each answer in descending order.

(a) $(2x-3)^2+(x+4)(2x^2-x+5)-(3x^3-7x^2+2x)$

(b) $(3a+2)^2-(a-1)(a^2+3a-4)$

(c) $(2m-5n)(2m+5n)+(2m-5n)^2$

(d) $(x+3)(x-3)(x^2+9)-(x^2-9)^2$

(e) In (c) and (d) a whole pair of terms vanished. In each case name the two terms that cancelled, and say which of the two techniques in the expression produced each of them.

6. Three factors at a time. In every one of these, *one* pair of the three factors is a special product — find that pair, multiply it first, and say in each case why that pairing was the short one.

(a) $(x-5)(x+5)(x-2)$ (b) $(2a+3)(2a+3)(a-4)$ (c) $(3p-q)(3p+q)(9p^2+q^2)$

(d) Expand $(y-4)^3$. State its degree and leading coefficient, and say how you could have predicted both without doing any of the work.

7. **One coefficient only.** Do *not* expand these completely — find only the number asked for, and show the partial products that feed into it.

(a) the coefficient of x^2 in $(x+3)(2x^2-5x+4)$

(b) the coefficient of x^3 in $(x^2-4)(3x^3+2x^2-x+7)$

(c) the constant term of $(2x-5)(x^2+3x-4)(x+1)$

(d) the degree and leading coefficient of $(3x^2-x)(2x^3+5)(x-4)$

(e) the coefficient of xy in $(2x+3y-1)^2$

(f) the coefficient of x in $(x-1)(x+4)(x-6)$

8. **Working backwards.** Every expansion below came from a product. Recover the missing factor, and check each answer by expanding.

(a) $(x+4)(_____) = x^3+4x^2-3x-12$

(b) $(2x-1)(_____) = 2x^3+5x^2-13x+5$

(c) $(_____)^2 = 25a^2-60ab+36b^2$

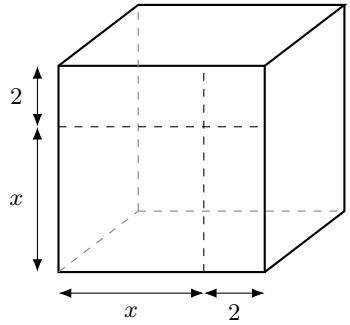
(d) (_____)(_____)
$= 49x^4 - 16y^2$

(e) $(2x - 3)$(_____)
$= 8x^3 - 27$

(f) (_____)$(x^2 - 5x + 6)$
$= x^3 - x^2 - 14x + 24$

Part C — Geometry and Applications

9. A cube has edge $(x + 2)$ cm. Each of its three edges is divided into a piece of length x and a piece of length 2, and the cube is sliced right through on all three cuts. This breaks it into 8 rectangular blocks.



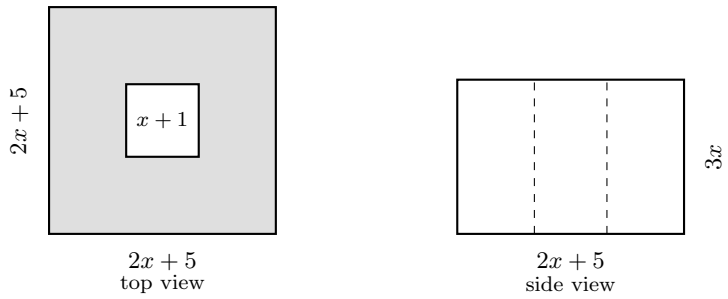
(a) **By algebra.** Expand $(x + 2)^2$, then multiply by $(x + 2)$ again to write the volume in descending order.

(b) **By the blocks.** Of the 8 pieces, one is x by x by x, three are x by x by 2, three are x by 2 by 2, and one is 2 by 2 by 2. Write each group's total volume, then add all four groups.

(c) Your two answers must be identical. Check them a third way at $x = 3$: state the real edge length, cube it, and confirm both the expansion and the four groups of blocks give that number.

(d) The coefficients of $(x + 2)^3$ are 1, 6, 12, 8. Say which group of blocks produces each one, and explain where the 3 in "three slabs" and "three bars" comes from geometrically.

10. A solid block is a rectangular prism with a square cross-section of side $(2x + 5)$ cm and height $3x$ cm. A square shaft of side $(x + 1)$ cm is drilled straight through it from top to bottom, as shown.



(a) **Cross-sectional area, way one.** Expand $(2x + 5)^2$ and $(x + 1)^2$ separately, then subtract and simplify.

(b) **Cross-sectional area, way two.** The same area is a difference of two squares. Apply $a^2 - b^2 = (a - b)(a + b)$ with $a = 2x + 5$ and $b = x + 1$, simplify both brackets, then expand. Confirm you get the same trinomial as in (a).

(c) Multiply by the height to write the volume of the drilled block in descending order.

(d) State the degree, leading coefficient and constant term of your volume, and say why the constant term had to be 0.

(e) Check at $x = 1$: give the three real measurements, work out the volume arithmetically, and compare with your polynomial.

(f) Which of the two routes in (a) and (b) would you rather use if the outer side were $(7x + 13)$ and the shaft were $(3x + 4)$? Say why, in terms of how many terms each route carries.

11. A rectangular plot measures $(x + 7)$ m long by $(x + 2)$ m wide. The owner shortens the **length** by k m and lengthens the **width** by the same k m, so the plot keeps the same total fencing but changes shape.

(a) Expand the original area. Then show that the perimeter really is unchanged by any k.

(b) Take $k = 3$. Write the two new side lengths, expand the new area, and state the change exactly.

(c) Check (b) at $x = 1$ and at $x = 5$ by working out all four real side lengths and both areas.

(d) Now keep k as a letter. Expand $(x + 7 - k)(x + 2 + k)$ in full, subtract the original area, and simplify the change to a product of two factors.

(e) Your answer to (d) contains no x — the change is the same whatever the plot's size. Use it to find the two values of k that leave the area *unchanged*, and explain what each one is doing to the rectangle.

12. A closed storage box has volume $(2x^3 + 11x^2 + 17x + 6)$ cm^3. Two of its three edges are known to be $(x + 2)$ cm and $(2x + 1)$ cm.

(a) Multiply the two known edges, then work backwards to find the third edge. Check by expanding all three factors.

(b) Using $SA = 2(lw + lh + wh)$, write and simplify an expression for the box's surface area in cm^2. Expand all three products before you double anything.

(c) State the degree, leading coefficient and constant term of the volume and of the surface area.

(d) Explain why the volume comes out one degree higher than the surface area, using the number of binomial factors in each.

(e) Check both at $x = 1$: give the three real edges, then the volume and the surface area two ways each.

Part D — Reasoning and Error Analysis

13. Each claim below is about the whole unit. Decide **true** or **false**, then justify: prove a true claim with an argument, and kill a false one with a counterexample.

(a) "The product of two binomials always has three terms."
Give *two* counterexamples: one product with fewer than three terms and one with more.

(b) "The degree of a product is the sum of the degrees of the two factors."
Argue this one carefully — your argument must say why the highest-degree term can never be cancelled by anything else.

(c) "The degree of a *sum* is the sum of the degrees."
Then state what the degree of a sum actually is, and when that rule fails.

(d) "A binomial times a trinomial always has exactly six terms."
Say what the six is really counting, give the largest number of terms that can survive, and give an example that ends with only two.

14. A student was asked to simplify $2(x-4)^2-(x+3)(x^2-2x+5)$ and wrote:

$2(x-4)^2-(x+3)(x^2-2x+5)$	
$=2(x^2-16)-(x^3+x^2-x+15)$	Line 1
$=2x^2-32-x^3-x^2+x+15$	Line 2
$=-x^3+x^2+x-17$	Line 3
"a polynomial of degree 3 with leading coefficient 1"	Line 4

There are **three** separate errors in this work.

(a) Substitute $x=1$ into the original expression and into Line 3 to show the answer is wrong.

(b) Name the three errors, one per line, and say which lesson's rule each one breaks.

(c) Redo the whole simplification correctly, in descending order.

(d) State the degree, leading coefficient and constant term of your answer, then check it at $x=1$ against your value from (a).

15. **Substitution as a check — and its limits.** A student claims

$$(x+1)(x^2+2x-3) = x^3+3x^2-3x-3.$$

(a) Expand the left side correctly and say which single coefficient the student got wrong.

(b) Substitute $x=0$ into the original product and into the student's answer. What do you get, and what would a student who checked only at $x=0$ conclude?

(c) Now check at $x=1$, $x=2$ and $x=-1$. Record all three pairs of values.

(d) Subtract the student's answer from the correct expansion. Use the polynomial you get to explain *exactly* which values of x make the two expressions agree — and why $x=0$ was the worst possible value to test.

(e) Explain why one agreeing substitution is weak evidence while three are strong. Then say honestly whether three agreements *prove* two cubic expressions equal, and what does.

Part E — Challenge

16. **Group before you expand.** Each of these becomes short if you first treat a whole bracketful of terms as a single a or b.

(a) $\big[(x+y)+3\big]\big[(x+y)-3\big]-(x-y)^2$

(b) $(a+b+c)(a+b-c)-(a-b)^2$

(c) $(x^2+x+1)(x^2-x+1)$
(Hint: group as $(x^2+1)+x$ and $(x^2+1)-x$.)

(d) Use your answer to (c), together with $(x-1)(x+1)$, to expand $(x-1)(x+1)(x^2+x+1)(x^2-x+1)$ in two short steps. Then check it at $x=2$.

17. **Three consecutive integers** can be written $n-1$, n and $n+1$.

(a) Expand $(n-1)(n+1)$ to prove that the product of the outer two is always exactly 1 less than the square of the middle one.

(b) Expand $(n-1)n(n+1)$ and state the result as a binomial in descending order. Give its degree and constant term.

(c) Use (b) to evaluate $19 \times 20 \times 21$ without a calculator. Show the substitution.

(d) Expand $(n-1)^2 + n^2 + (n+1)^2$ and simplify. Which terms cancelled, and why?

(e) Use (d) to evaluate $29^2 + 30^2 + 31^2$ mentally, then verify by squaring all three.

(f) Expand $(n+1)^3 - n^3$ and use the result to find $25^3 - 24^3$ without cubing anything.

18. **An identity worth keeping.** Consider $(x^2 + ax + b)(x^2 - ax + b)$, where a and b are numbers.

(a) Expand it fully and show that every odd-degree term disappears, leaving $x^4 + (2b - a^2)x^2 + b^2$.
(Hint: group as $(x^2 + b) + ax$ and $(x^2 + b) - ax$.)

(b) Choose a and b so that the x^2 term vanishes as well and the product is exactly $x^4 + 4$. Write out the two factors you get.

(c) Test your factors from (b) at $x = 3$ and again at $x = 5$: evaluate $x^4 + 4$ and evaluate the product of the two factors, and confirm they match both times.

(d) Repeat the choice in (b) to write $x^4 + 64$ as a product of two trinomials. State the general rule connecting a, b and the constant.

19. *(Multiple Choice)* Expand and simplify $(2x - 3)(x^2 + 4x - 1)$.

A. $2x^3 + 5x^2 - 14x + 3$
B. $2x^3 + 5x^2 - 14x - 3$
C. $2x^3 + 11x^2 - 14x + 3$
D. $2x^3 + 5x^2 + 10x + 3$

20. *(Numerical Response)* The expression $(2x + 3)^2 - (x - 4)(x + 4) + (x - 1)(x^2 + x + 1)$ simplifies to $x^3 + ax^2 + bx + c$. The value of $a + b + c$ is _____.
(Record your answer as a three-digit number.)